CLA-3 – MACHINE LEARNING CASE STUDY

IRIS FLOWER SPECIES CLASSIFICATION USING K-NEAREST NEIGHBORS (KNN)

Case study title	Iris Flower Species Classification Using K-Nearest Neighbors (KNN)
Base paper	R. A. Fisher, “The Use of Multiple Measurements in Taxonomic Problems,” 1936.
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1. Introduction and Problem Statement

Classification is a supervised machine learning task in which a model learns from labelled examples and assigns a class to new observations. This case study uses measurements of Iris flowers to predict their species. Correct identification from measurable characteristics is a simple example of how data-driven classification can support biological identification and demonstrate the basic machine learning workflow.

The Iris dataset contains four measurements for each flower: sepal length, sepal width, petal length and petal width, all measured in centimetres. The target has three classes: Iris setosa, Iris versicolor and Iris virginica. The problem addressed is: Can a K-Nearest Neighbors classifier use these four measurements to predict the species of an unseen flower? This is a three-class classification problem, not a regression problem.

2. Base Paper Summary

The base paper is R. A. Fisher’s 1936 study, “The Use of Multiple Measurements in Taxonomic Problems.” Fisher examined how multiple measurements could be combined to distinguish groups in a taxonomic setting. The paper discusses measurements of Iris flowers and the use of statistical functions to separate species. It established an influential example of using quantitative measurements for classification and discrimination.

The present implementation does not reproduce Fisher’s discriminant-function method. Instead, it uses the same general Iris measurement problem to demonstrate K-Nearest Neighbors, a distance-based machine learning classifier. The paper provides the historical and problem context; the accuracy and other results in this report come only from the KNN experiment executed for this case study.

3. Method / Model Explanation

K-Nearest Neighbors (KNN) is a supervised, instance-based learning algorithm. It stores the training examples. To classify a new flower, it calculates the distance from that flower to every training example, selects the k closest examples, and assigns the class that appears most often among those neighbours. Here, k = 5.

$$\mathbf{d(x, z) = \sqrt{[\ \Sigma_{i=1}^{n} (x_i - z_i)^2\ ]}}$$

In this equation, x is the new flower’s feature vector, z is a training flower’s feature vector, x_i and z_i are their values for feature i, and n is the number of features (four). The model uses Euclidean distance. Before distance calculation, StandardScaler standardises each feature using the training data so that features with larger numeric scales do not dominate the distance.

Steps: (1) load the labelled Iris dataset; (2) separate the four input features (X) from the species labels (y); (3) split the data into 80% training and 20% testing sets using stratification; (4) standardise features; (5) train KNN with five neighbours; (6) predict labels for the test set; and (7) evaluate predictions.

Worked example: suppose the five nearest training flowers to a new sample have labels setosa, setosa, versicolor, setosa and virginica. The majority label is setosa (3 out of 5), so KNN predicts Iris setosa. This illustrates the voting rule; the actual program calculates distances from the supplied measurements rather than manually selecting neighbours.

4. Implementation

The implementation uses Python and Scikit-learn. The dataset is loaded directly using sklearn.datasets.load_iris(), so a separate dataset download is not required. Scikit-learn documents this as a classic multi-class dataset containing 150 samples, 50 per class, four numeric features and three classes.

The experiment uses an 80:20 train-test split, stratified by species, with random_state=42 for reproducibility. The training set therefore contains 120 examples and the test set contains 30. A pipeline applies StandardScaler followed by KNeighborsClassifier(n_neighbors=5). The pipeline is fitted only on the training data and then used to predict the test labels.

The executable source is provided as iris_knn.py and as a Jupyter notebook. To run it, install Scikit-learn using “pip install scikit-learn” and execute “python iris_knn.py”, or open the notebook in Google Colab/Jupyter and run all cells. A common issue with KNN is that measurements on different scales can distort distance calculations; StandardScaler addresses this. Another important precaution is to fit scaling only on training data, which the pipeline handles.

Code source: Scikit-learn documentation and API for the Iris dataset, train-test splitting, scaling and KNN. Project code is included with this report.

5. Results and Discussion

The executed model was evaluated on 30 unseen test examples. It correctly classified 28 examples and misclassified 2. The resulting test accuracy was 93.33%.

Metric	Setosa	Versicolor	Virginica
Precision	1.00	0.83	1.00
Recall	1.00	1.00	0.80
F1-Score	1.00	0.91	0.89

Confusion matrix (rows = actual species; columns = predicted species):

Actual / Predicted	Setosa	Versicolor	Virginica
Setosa	10	0	0
Versicolor	0	10	0
Virginica	0	2	8

Correct example: measurements [4.4, 3.0, 1.3, 0.2] were classified as setosa; the actual label was setosa.

Incorrect example: measurements [6.0, 3.0, 4.8, 1.8] were predicted as versicolor, while the actual label was virginica. Similar measurements between species can cause KNN to select neighbours from another class.

The result indicates that flower measurements contain useful information for distinguishing the three species. However, this is a small, clean, well-known dataset. Performance on it does not guarantee similar performance on noisy measurements, other Iris populations or unrelated flower species. Results can also change with the train-test split, k value, distance metric and scaling choices.

6. Conclusion

This case study demonstrated a complete supervised classification workflow using the Iris dataset and KNN. With four flower measurements and five nearest neighbours, the model achieved 93.33% accuracy on the held-out 30-sample test set. The experiment shows that KNN can classify Iris species effectively on this dataset. The model is simple and easy to implement, but its performance depends on feature scaling, the choice of k and the representativeness of the data.

7. References

[1] R. A. Fisher, “The Use of Multiple Measurements in Taxonomic Problems,” Annals of Eugenics, vol. 7, no. 2, pp. 179–188, 1936, doi: 10.1111/j.1469-1809.1936.tb02137.x.

[2] Scikit-learn Developers, “The Iris dataset,” Scikit-learn documentation. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_iris.html. Accessed: Nov. 8, 2026.

[3] Scikit-learn Developers, “KNeighborsClassifier,” Scikit-learn documentation. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html. Accessed: Nov. 8, 2026.

[4] Scikit-learn Developers, “StandardScaler,” Scikit-learn documentation. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html. Accessed: Nov. 8, 2026.

8. Declaration

I confirm that this case study is my own work and that all results reported were obtained by me.

Signature: _____

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